

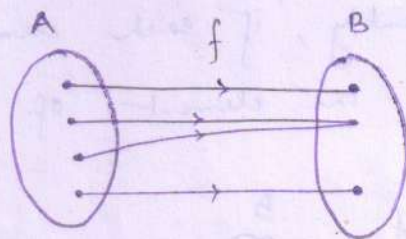
Defⁿ:

Let A and B be two non-empty sets. A function f from A to B is a set of ordered pairs $f \subseteq A \times B$ with the property that for each element x in A there is a unique element y in B such that $(x, y) \in f$.

The statement " f is a function from A to B " is usually represented symbolically by

$$f: A \rightarrow B \quad \text{or} \quad A \xrightarrow{f} B.$$

A function can be represented pictorially as shown in the figure below:



Note: i) There may be some elements of the set B which are not associated to any element of the set A .

ii) Each element of the set A must be associated to one and only one element of the set B .

If f is a function from A to B , then A is called the domain of f denoted by $\text{dom } f$, its members are the first co-ordinates of the ordered pairs belonging to f and the set B is called the co-domain.

If $(x, y) \in f$, it is customary to write $y = f(x)$, y is called the image of x ; and x is a pre-image of y . y is also called the value of f at x . The set consisting

of all the images of the elements of A under the function f ② is called the range of f . It is denoted by $f(A)$.

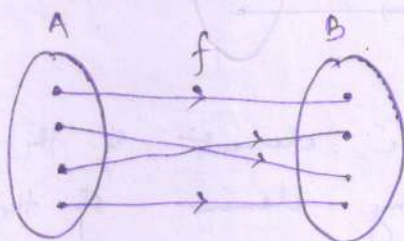
Thus range of $f = \{f(x) : \text{for all } x \in A\}$.

Note: Range of f is a subset of B (Co-domain) which may or may not be equal to B .

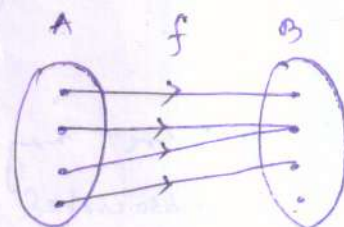
Types of functions:

1) One-to-one function: A function from A into B is one-to-one or injective, if for all elements x_1, x_2 in A such that $f(x_1) = f(x_2)$ implies $x_1 = x_2$.

That is, if no elements of A are assigned to the same element in B or equivalently, if each element of the range corresponds to the exactly one element of the domain, then f is one-to-one.



Injective



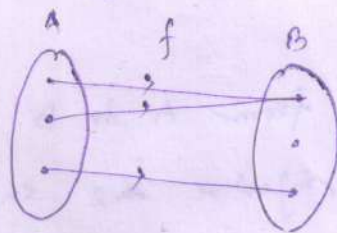
Not injective

Ex: Let $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$ and let $f(1) = a$, $f(2) = c$ and $f(3) = d$. Then f is injective since the different elements 1, 2, 3 in A are assigned to the different elements a, c, d respectively in B .

Ex: $f(x) = 3x - 1$ is one-to-one function because

$$f(x_1) = f(x_2) \Rightarrow 3x_1 - 1 = 3x_2 - 1 \Rightarrow x_1 = x_2.$$

27) Many-one function: A function f from A to B is said to be many-one if and only if two or more elements of A have same image in B . (3)



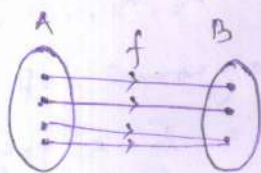
Many-one function

Ex: Let $f(x) = x^2$, x is any real number and $f: \mathbb{R} \rightarrow \mathbb{R}$, then f is many-one function. For $x=1$, $f(1) = 1^2 = 1$ and $x=-1$, $f(-1) = (-1)^2 = 1$. Thus we find $f(1) = f(-1) = 1$ which shows that two distinct numbers -1 and 1 are assigned to the same number 1 under f . Therefore, f is many-one function.

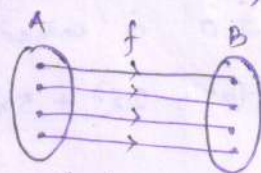
3) Into function: A function f from A to B is called into function if and only if there exists at least one element in B which is not the image of any element in A , i.e., the range of f is a proper subset of co-domain of f .

4) Onto function: A function f from A to B is onto, or surjective if every element of B is the image of some element in A , that is, if $B = \text{range of } f$.

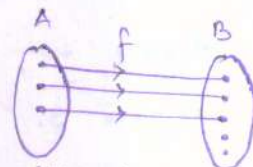
In order to check whether $y = f(x)$ from a set A to set B is onto or not, write x in terms of y and see if for every $y \in B$, $x \in A$. If so, it is onto. Otherwise, it is into.



Surjective



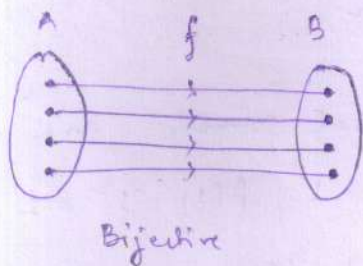
Surjective



Not surjective

Ex: Let $f(x) = x^2$, x is any real number and $f: \mathbb{R} \rightarrow \mathbb{R}$, then f is not onto. The reason is that we cannot find a real number whose square is negative, then the range of f cannot be equal to $\mathbb{R}$. (4)

5/ Bijjective Function: A function f from A to B is said to be bijective if f is both injective and surjective i.e., both one-to-one and onto.



Classification of functions:

Functions can be classified mainly into two groups

1) Algebraic function: A function which consists of a finite number of terms involving powers and roots of the independent variable x and the four fundamental operations of addition, subtraction, multiplication and division is called algebraic function.

Three particular cases of algebraic functions are:

1) Polynomial functions:

A function of the form $a_0x^n + a_1x^{n-1} + \dots + a_n$ where n is a positive integer and $a_0, a_1, \dots, a_n$ are real constants and $a_0 \neq 0$ is called a polynomial of x in degree n . e.g., $f(x) = 2x^3 + 5x^2 + 7x - 3$ is a polynomial of degree 3.

i) Rational functions: A function of the form $\frac{f(x)}{g(x)}$ where

$f(x)$ and $g(x)$ are polynomials in x ,
 $g(x) \neq 0$ is called a rational function, e.g., $F(x) = \frac{x^2 + 2x + 1}{x + 2}$.

ii) Irrational functions: The functions involving radicals are called irrational functions.

$f(x) = \sqrt[3]{x} + 5$ is an irrational function.

2) Transcendental functions: A function which is not algebraic is called transcendental function.

i) Trigonometric functions: The six functions $\sin x$, $\cos x$, $\tan x$, $\sec x$, $\operatorname{cosec} x$, $\cot x$ where the angle x is measured in radian are called trigonometric functions.

ii) Inverse trigonometric functions: The six functions $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$, $\cot^{-1} x$, $\sec^{-1} x$, $\operatorname{cosec}^{-1} x$ are called inverse trigonometric functions.

iii) Exponential functions: A function $f(x) = a^x$ ($a > 0$) satisfying the law $a^x \cdot a^y = a^{x+y}$ is called the exponential function.

iv) Logarithm functions: The inverse of the exponential function is called the logarithm function.

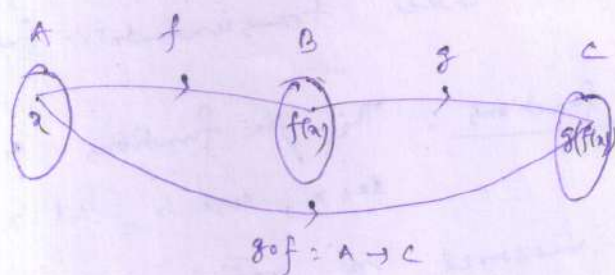
So, if $y = a^x$ ($a > 0, a \neq 1, x \in \mathbb{R}, y > 0$), then $x = \log_a y$ is called logarithm function.

Composition of functions:

(6)

Let $f: A \rightarrow B$ and $g: B \rightarrow C$. The composition of f and g , denoted by $g \circ f$, read as 'g of f' results in a new function from A to C and is given by $(g \circ f)(x) = g(f(x))$ for all x in A . Hence, the composition $g \circ f$ first applies f to map A into B , and it then employs g to map B to C . In other words, the range space of f becomes the domain space of g .

The figure below illustrates the composition of two functions f and g .



Ex: Let $A = \{1, 2, 3\}$, $B = \{a, b\}$ and $C = \{r, s\}$ and $f: A \rightarrow B$ be defined by $f(1) = a$, $f(2) = a$, $f(3) = b$ and $g: B \rightarrow C$ be defined by $g(a) = s$, $g(b) = r$.

Then $g \circ f: A \rightarrow C$ is defined by

$$(g \circ f)(1) = g(f(1)) = g(a) = s$$

$$(g \circ f)(2) = g(f(2)) = g(a) = s$$

$$(g \circ f)(3) = g(f(3)) = g(b) = r$$

Ex: If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ are defined by the formulas $f(x) = x + 2$ for all x in $\mathbb{R}$ and $g(x) = x^2$ for all x in $\mathbb{R}$.

$$\text{Then } (g \circ f)(x) = g(f(x)) = g(x + 2) = (x + 2)^2 = x^2 + 4x + 4$$

$$(f \circ g)(x) = f(g(x)) = f(x^2) = x^2 + 2$$

Note: $g \circ f \neq f \circ g$, since, for ex,

(7)

$$(g \circ f)(1) = g(f(1)) = g(1+2) = g(3) = 3^2 = 9, \text{ while}$$

$$(f \circ g)(1) = f(g(1)) = f(1^2) = f(1) = 1+2 = 3.$$

Thus the composition of function is not commutative.

However, the associative law is true for functions under the operation of compositions.

Theorem: (Associative law of Function Composition)

Let $f: A \rightarrow B$, $g: B \rightarrow C$ and $h: C \rightarrow D$,
then $h \circ (g \circ f) = (h \circ g) \circ f$.

→

Since, $f: A \rightarrow B$, $g: B \rightarrow C$ and $h: C \rightarrow D$,

then $g \circ f: A \rightarrow C$ and $h \circ g: B \rightarrow D$

Hence, $h \circ (g \circ f): A \rightarrow D$ and $(h \circ g) \circ f: A \rightarrow D$.

So, $\text{dom}[h \circ (g \circ f)] = \text{dom}[(h \circ g) \circ f]$

Let $x \in A$, $y \in B$, $z \in C$ such that $f(x) = y$ and $g(y) = z$.

$$\begin{aligned} \text{Then, } [(h \circ g) \circ f](x) &= (h \circ g)[f(x)] \\ &= (h \circ g)(y) = h[g(y)] = h(z) \quad \dots (i) \end{aligned}$$

$$\begin{aligned} \text{Also, } [h \circ (g \circ f)](x) &= h[(g \circ f)(x)] = h[g(f(x))] \\ &= h[g(y)] = h(z) \quad \dots (ii) \end{aligned}$$

From (i) and (ii), we get that

$$[(h \circ g) \circ f](x) = [h \circ (g \circ f)](x) \text{ for all } x \text{ in } A$$

$$\Rightarrow (h \circ g) \circ f = h \circ (g \circ f) \quad (\text{proved})$$

Theorem: Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions

- If f and g are injections then $g \circ f: A \rightarrow C$ is an injection.
- If f and g are surjections then so is $g \circ f$.
- If f and g are bijections then so is $g \circ f$.

→
a) Let $a_1, a_2 \in A$.

By definition of composition, we have,

$$(g \circ f)(a_1) = (g \circ f)(a_2) \Rightarrow g(f(a_1)) = g(f(a_2))$$

$$\Rightarrow f(a_1) = f(a_2), \text{ since } g \text{ is injective}$$

$$\Rightarrow a_1 = a_2, \text{ since } f \text{ is injective}$$

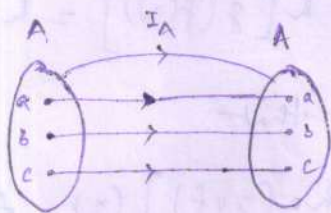
$\therefore g \circ f$ is injective.

b) Let $c \in C$. Then we can find an element $a \in A$ such that $(g \circ f)(a) = c$. Since g is onto C , there is an element $b \in B$ such that $g(b) = c$. Then, since f is onto B , there exists $a \in A$ such that $f(a) = b$. Thus, $(g \circ f)(a) = g(f(a)) = g(b) = c$.

c) This part follows immediately from the preceding parts.

(proved)

Identity Function: The function $f: A \rightarrow A$ defined by $f(x) = x$ for every $x \in A$ is called the identity of A and is denoted by I_A .



Theorem: The composition of any function with the identity function is the function itself, i.e.,

$$(f \circ I_A)(x) = (I_B \circ f)(x) = f(x).$$

Proof:

Let A and B be two non-empty sets.

Let I_A be the identity function on A and let $f: A \rightarrow B$.

Let $x \in A$ and $y \in B$, so that $y = f(x)$.

So, $I_A: A \rightarrow A$, i.e., $I_A(x) = x$, for all x in A .

By the definition of the composite function $f \circ I_A: A \rightarrow B$, then

$$(f \circ I_A)(x) = f[I_A(x)] = f(x) \quad \text{--- (i)}$$

Again, let $f: A \rightarrow B$ and I_B be the identity function on B .

So, $I_B: B \rightarrow B$, i.e., $I_B(y) = y$, for all y in B .

Then, $(I_B \circ f)(x) = I_B[f(x)] = I_B(y) = y = f(x) \quad \text{--- (ii)}$

From (i) and (ii), we get,

$$(f \circ I_A)(x) = (I_B \circ f)(x) = f(x)$$

(proved)

Note: In particular, if $f: A \rightarrow A$, then $f \circ I_A = I_A \circ f = f$.

Inverse of a function:

Let $f: A \rightarrow B$

A map $g: B \rightarrow A$ is called the inverse of f if $g \circ f = I_A$ and $f \circ g = I_B$ i.e.,

$$g[f(x)] = x \text{ for all } x \in A$$

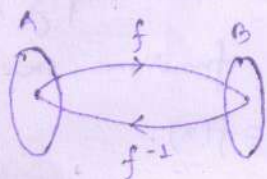
$$\text{and } f[g(y)] = y \text{ for all } y \in B$$

Thus, if $f(x) = y$ then $g(y) = g[f(x)] = x$.

The inverse g of f is denoted by f^{-1} .

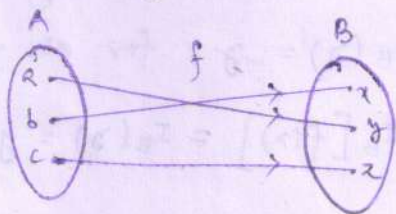
$$\text{Thus } f(x) = y \Leftrightarrow x = f^{-1}(y).$$

Note: A necessary and sufficient condition for $f: A \rightarrow B$ to have the inverse of $f^{-1}: B \rightarrow A$ is that f be bijective. Then f takes x to y and f^{-1} takes y to x . Thus for a bijective f , f^{-1} is obtained by 'reversing' arrows.

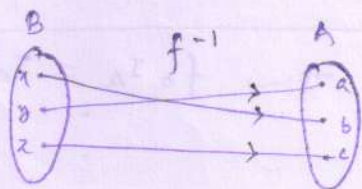


The concept of inverse function is illustrated in the following example.

Ex: Let the function $f: A \rightarrow B$ be defined by the diagram as shown in the figure below.



Then f is one to one and onto. Therefore f^{-1} , the inverse function, exists. One can describe $f^{-1}: B \rightarrow A$ by the diagram as shown in the figure below.



Ex: Show that the function $f(x) = x^3$ and $g(x) = x^{1/3}$ for all $x \in \mathbb{R}$ are inverses of one another.

→ Since, $(f \circ g)(x) = f(x^{1/3}) = x = Ix$

and $(g \circ f)(x) = g(x^3) = x = Ix$

then $f = g^{-1}$ or $g = f^{-1}$.

Ex: Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d\}$ and let $f = \{(1, a), (2, a), (3, d), (4, c)\}$. Show that f is a function but f^{-1} is not a function. (11)

→ Here we have $f(1) = a$, $f(2) = a$, $f(3) = d$ and $f(4) = c$.

Since each set $f(n)$ is a single value, f is a function.

Now, $f^{-1} = \{(a, 1), (a, 2), (d, 3), (c, 4)\}$ but f^{-1} is not a function,

since $f^{-1}(a) = \{1, 2\}$.

Ex: Show that the mapping $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = ax + b$, where $a, b, x \in \mathbb{R}$, $a \neq 0$ is invertible. Define its inverse.

→ For, if $x_1, x_2 \in \mathbb{R}$, then

$$f(x_1) = f(x_2) \Rightarrow ax_1 + b = ax_2 + b$$

$$\Rightarrow ax_1 = ax_2$$

$$\Rightarrow x_1 = x_2$$

This proves f is one-to-one.

Again, if $y \in \mathbb{R}$

$$y = f(x) \Rightarrow y = ax + b$$

$$\Rightarrow x = \frac{(y-b)}{a}$$

Thus for $x \in \mathbb{R}$, there exists $\frac{y-b}{a} \in \mathbb{R}$ such that

$$f\left(\frac{1}{a}(y-b)\right) = a\left(\frac{1}{a}(y-b)\right) + b = y - b + b = y \quad \Rightarrow$$

Hence f is one-to-one and onto therefore f^{-1} exists and it is defined by

$$f^{-1}(y) = \frac{1}{a}(y-b).$$

Theorem: If a mapping $f: A \rightarrow B$ is one-to-one and onto, then prove that (12)
inverse mapping $f^{-1}: B \rightarrow A$ is also one-to-one and onto.

→ Here $f: A \rightarrow B$ is one-to-one and onto

$a_1, a_2 \in A$ and $b_1, b_2 \in B$ so that

$$b_1 = f(a_1), b_2 = f(a_2)$$

$$\text{and } a_1 = f^{-1}(b_1), a_2 = f^{-1}(b_2)$$

As f is one to one.

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$$

$$\text{or, } b_1 = b_2 \Leftrightarrow f^{-1}(b_1) = f^{-1}(b_2)$$

$$\text{ie, } f^{-1}(b_1) = f^{-1}(b_2) \Rightarrow b_1 = b_2$$

$\therefore f^{-1}$ is one to one function

As f is onto

every element of B is associated with a unique element of A

ie, for any $a \in A$ is pre-image of some $b \in B$ where $b = f(a)$

$$\Rightarrow a = f^{-1}(b)$$

ie, for $b \in B$, there exists f^{-1} image $a \in A$

Hence f^{-1} is onto.

Theorem: If $f: A \rightarrow B$ is a one to one and onto function, then

$$\text{a) } f^{-1} \circ f = I_A \text{ and b) } f \circ f^{-1} = I_B$$

where I_A and I_B are identity mapping on set A and set B .

→ Since $f: A \rightarrow B$ is one to one and onto, f^{-1} exists

For all $a \in A, b \in B$, we have,

$$b = f(a), a = f^{-1}(b)$$

$$\Rightarrow b = f(f^{-1}(b))$$

$$= (f \circ f^{-1})(b) \text{ for all } b \in B$$

$$\therefore I_B = f \circ f^{-1}$$

Again

$$a = f^{-1}(b)$$

$$= f^{-1}(f(a)) \text{ for all } a \in A$$

$$= (f^{-1} \circ f)(a)$$

$$\therefore I_A = f^{-1} \circ f$$

Theorem: If $f: A \rightarrow B$ and $g: B \rightarrow C$ be one to one ~~to~~ onto functions, then $g \circ f$ is also one-to-one ~~to~~ onto and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. (13)

→ Since f is one-to-one, $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ for $x_1, x_2 \in A$.

Again since g is one-to-one, $g(y_1) = g(y_2) \Rightarrow y_1 = y_2$ for $y_1, y_2 \in B$.

Now, $g \circ f$ is one-to-one, since $(g \circ f)(x_1) = (g \circ f)(x_2)$

$$\Rightarrow g[f(x_1)] = g[f(x_2)]$$

$$\Rightarrow f(x_1) = f(x_2) \quad (g \text{ is one to one})$$

$$\Rightarrow x_1 = x_2 \quad (f \text{ is one to one})$$

Since g is onto, for $z \in C$, there exists $y \in B$ such that $g(y) = z$. Also f being onto there exists $x \in A$ such that $f(x) = y$.

$$\text{Hence } z = g(y) = g[f(x)] = (g \circ f)(x).$$

This shows that every element $z \in C$ has pre image under $g \circ f$. So, $g \circ f$ is onto.

Thus $g \circ f$ is one-to-one and onto function and hence

$(g \circ f)^{-1}$ exists.

By the defⁿ of composite functions, $g \circ f: A \rightarrow C$. So $(g \circ f)^{-1}: C \rightarrow A$.

Also $g^{-1}: C \rightarrow B$ and $f^{-1}: B \rightarrow A$.

Then by defⁿ of composite functions $f^{-1} \circ g^{-1}: C \rightarrow A$.

$\therefore$ Domain of $(g \circ f)^{-1} = \text{domain of } f^{-1} \circ g^{-1}$.

$$\text{Now, } (g \circ f)^{-1}(z) = x \Leftrightarrow (g \circ f)(x) = z$$

$$\Leftrightarrow g(f(x)) = z$$

$$\Leftrightarrow g(y) = z \text{ where } y = f(x)$$

$$\Leftrightarrow y = g^{-1}(z)$$

$$\Leftrightarrow f^{-1}(y) = f^{-1}(g^{-1}(z))$$

$$= (f^{-1} \circ g^{-1})(z)$$

$$\Leftrightarrow x = (f^{-1} \circ g^{-1})(z) \quad [\because f^{-1}(y) = x]$$

Thus $(g \circ f)^{-1}(z) = (f^{-1} \circ g^{-1})(z)$. So $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ (proved)

Ex: let f and g be functions from the integers to the integers defined by $f(n) = n^2$, $g(n) = 2^n$. (14)

find $f \circ f$, $g \circ g$, $f \circ g$, $g \circ f$.

$$\begin{aligned} (f \circ f)(n) &= f[f(n)] = f(n^2) = (n^2)^2 = n^4 \\ (g \circ g)(n) &= g[g(n)] = g(2^n) = (2)^{2^n} = 2^{2^n} \\ (f \circ g)(n) &= f[g(n)] = f(2^n) = (2^n)^2 = 2^{2n} = 4^n \\ (g \circ f)(n) &= g[f(n)] = g(n^2) = 2^{n^2} \end{aligned}$$

Ex: if function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 3x-4 & x > 0 \\ -3x+2 & x \leq 0 \end{cases}$$

Determine a) $f(0)$, $f(2/3)$, $f(-2)$

b) $f^{-1}(0)$, $f^{-1}(2)$, $f^{-1}(-7)$

$$\rightarrow a) f(0) = -3(0) + 2 = 2$$

$$f(2/3) = 3(2/3) - 4 = -2$$

$$f(-2) = -3(-2) + 2 = 8$$

$$\begin{aligned} b) f^{-1}(0) &= \{x \in \mathbb{R} \mid f(x) = 0\} \\ &= \{x \in \mathbb{R} \mid x > 0 \text{ and } 3x-4=0\} \cup \{x \in \mathbb{R} \mid x \leq 0 \text{ and } -3x+2=0\} \\ &= \{x \in \mathbb{R} \mid x > 0 \text{ and } x = 4/3\} \cup \{x \in \mathbb{R} \mid x \leq 0 \text{ and } x = 2/3\} \\ &= \{4/3\} \cup \emptyset = \{4/3\} \end{aligned}$$

$$\begin{aligned} f^{-1}(2) &= \{x \in \mathbb{R} \mid f(x) = 2\} \\ &= \{x \in \mathbb{R} \mid x > 0 \text{ and } 3x-4=2\} \cup \{x \in \mathbb{R} \mid x \leq 0 \text{ and } -3x+2=2\} \\ &= \{x \in \mathbb{R} \mid x > 0 \text{ and } x=2\} \cup \{x \in \mathbb{R} \mid x \leq 0 \text{ and } x=0\} \\ &= \{2\} \cup \{0\} = \{0, 2\} \end{aligned}$$

$$\begin{aligned} f^{-1}(-7) &= \{x \in \mathbb{R} \mid x > 0 \text{ and } 3x-4=-7\} \cup \{x \in \mathbb{R} \mid x \leq 0 \text{ and } -3x+2=-7\} \\ &= \{x \in \mathbb{R} \mid x > 0 \text{ and } x=-1\} \cup \{x \in \mathbb{R} \mid x \leq 0 \text{ and } x=3\} \\ &= \emptyset \cup \emptyset = \emptyset \end{aligned}$$

Ex. Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 3x-1 & \text{for } x > 3 \\ 2x^2+3 & \text{for } -2 < x \leq 3 \\ 3x^2-7 & \text{for } x \leq -2 \end{cases}$$

(15)

Find out $f^{-1}(5)$.

$$\begin{aligned} \longrightarrow f^{-1}(5) &= \{x \in \mathbb{R} \mid f(x) = 5\} \\ &= \{x \in \mathbb{R} \mid x > 3 \text{ and } 3x-1=5\} \cup \{x \in \mathbb{R} \mid -2 < x \leq 3 \text{ and } 2x^2+3=5\} \\ &\quad \cup \{x \in \mathbb{R} \mid x \leq -2 \text{ and } 3x^2-7=5\} \\ &= \{x \in \mathbb{R} \mid x > 3 \text{ and } x=2\} \cup \{x \in \mathbb{R} \mid -2 < x \leq 3 \text{ and } x=\pm 1\} \\ &\quad \cup \{x \in \mathbb{R} \mid x \leq -2 \text{ and } x=\pm 2\} \\ &= \emptyset \cup \{-1, 1\} \cup \{-2\} = \{-2, -1, 1\}. \end{aligned}$$

Ex: Let $A=B=C=\mathbb{R}$. Consider the ^{bijjective} functions $f: A \rightarrow B$ and $g: B \rightarrow C$ defined by $f(a) = 2a+1$, $g(b) = \frac{b}{3}$. Verify $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

$\longrightarrow$ The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(a) = 2a+1$ is bijective. Hence f^{-1} exists.

$$\text{Let } b = f(a) = 2a+1 \quad \therefore a = \frac{b-1}{2}.$$

$$\text{Also } f(a) = b \Rightarrow a = f^{-1}(b) \quad \therefore \frac{b-1}{2} = f^{-1}(b) \quad \text{--- (1)}$$

Also, the functioning $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(b) = \frac{b}{3}$ is bijective. Hence g^{-1} exists.

$$\text{Let } c = g(b) = \frac{b}{3} \quad \therefore b = 3c.$$

$$\text{Also } g(b) = c \Rightarrow b = g^{-1}(c) \quad \therefore 3c = g^{-1}(c) \quad \text{--- (2)}$$

Now, $g \circ f: A \rightarrow C$ is defined by $(g \circ f)(a) = c$.

$$\therefore g(f(a)) = c \Rightarrow g(2a+1) = c$$

$$\therefore \frac{2a+1}{3} = c \Rightarrow a = \frac{3c-1}{2}$$

$$\text{Also, } (g \circ f)(a) = c \Rightarrow a = (g \circ f)^{-1}(c)$$

$$\text{i.e., } \frac{3c-1}{2} = (g \circ f)^{-1}(c)$$

$$\text{Again, } (f^{-1} \circ g^{-1})(c) = f^{-1}(g^{-1}(c)) = f^{-1}(3c) = \frac{3c-1}{2} \quad (\text{using (2) \& (1)})$$

$$\therefore (g \circ f)^{-1}(c) = (f^{-1} \circ g^{-1})(c) = \frac{3c-1}{2}$$

Hence, $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ is verified.

Ex: Let $f: X \rightarrow Y$ be an everywhere defined invertible function and A and B be arbitrary non-empty subsets of Y . Show that (16)

$$i) f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$$

$$ii) f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

→ i) Let $x \in X$, then

$$x \in f^{-1}(A \cup B) \Leftrightarrow f(x) \in \{A \cup B\}$$

$$\Leftrightarrow f(x) \in A \text{ or } f(x) \in B$$

$$\Leftrightarrow x \in f^{-1}(A) \text{ or } x \in f^{-1}(B)$$

$$\Leftrightarrow x \in \{f^{-1}(A) \cup f^{-1}(B)\}$$

$$\therefore f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$$

$$ii) x \in f^{-1}(A \cap B) \Leftrightarrow f(x) \in \{A \cap B\}$$

$$\Leftrightarrow f(x) \in A \text{ and } f(x) \in B$$

$$\Leftrightarrow x \in f^{-1}(A) \text{ and } x \in f^{-1}(B)$$

$$\Leftrightarrow x \in \{f^{-1}(A) \cap f^{-1}(B)\}$$

$$\therefore f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

Recursively Defined function

Recursion is defined a process of defining an object in terms of itself.

Total and partial functions:

Let N^n be the set of all n -tuples of elements of N , then any function $f: N^n \rightarrow N$ is called total function.

For ex, if $f: N \times N \rightarrow N$ is defined by

$$f(x, y) = x + y, \text{ then } f \text{ is a total function since it is defined for all } x, y \in N.$$

Any function $f: A \rightarrow N$ where $A \subseteq N^n$ is called partial function.

For ex, if $g: N \times N \rightarrow N$ is defined by $g(x, y) = x - y$, then g is called partial function since it is defined for only those $x, y \in N$ for which $x > y$.

Initial function:

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The initial functions are a set of primitive recursive functions which are accepted as self-evidently computable functions.

The initial functions over a set of natural numbers ($\mathbb{N}$) is given by

1) Zero function: $Z(x) = 0$ for all x .

2) Successor function: $S(x) = x+1$ for all x .

3) Projection function: $U_k^n(x_1, x_2, \dots, x_k, \dots, x_n) = x_k$ for $1 \leq k \leq n$

where n stands for the no. of arguments and k represents the selected arguments. Thus the projection selects one of the arguments from the lists.

Using the initial functions one can build other more complex primitive recursive functions by applying the following rules:

Composition: Let $f_1, f_2, \dots, f_n$ each be partial function of k variables and let g be a partial function of n variables. Then the composition of g with $f_1, f_2, \dots, f_n$ produces a partial function h of k variables given by

$$h(x_1, x_2, \dots, x_k) = g(f_1(x_1, x_2, \dots, x_k), \dots, f_n(x_1, x_2, \dots, x_k))$$

Primitive Recursion: Let g and h be primitive recursive function with n and $n+2$ variables respectively, then the $n+1$ variable function f defined by

$$1) f(x_1, x_2, \dots, x_n, 0) = g(x_1, x_2, \dots, x_n)$$

$$2) f(x_1, x_2, \dots, x_n, y+1) = h(x_1, x_2, \dots, x_n, y, f(x_1, x_2, \dots, x_n, y))$$

is also primitive recursive. Hence the variable y is called the recursive variable.

Def^m: A function f is called primitive recursive if and only if it can be constructed from the initial function by a finite no. of operation of composition and recursion.

Ex: Let $f_1(x, y) = x + y$, $f_2(x, y) = 3x$, $f_3(x, y) = xy$ and $g(x, y, z) = x + y + z$ be function over N . Then

(18)

$$g(f_1(x, y), f_2(x, y), f_3(x, y)) = g(x + y, 3x, xy) \\ = x + y + 3x + xy.$$

This gives the composition of g with f_1, f_2, f_3 .

Ex: Consider $f(0, x) = g(x)$, then the primitive recursion for $f(0+1, x) = h(0, x, f(0, x))$

Ex: Show that function $f(x, y) = x + y$ is primitive recursive function. Hence compute the value of $f(2, 4)$

→ Given that $f(x, y) = x + y$, using the defⁿ of recursion.

Here f is a function of two variables. If we want f to be defined by recursion, we need a function g of a single variable and a function h of three variables.

This can be written as

$$f(x, y+1) = x + (y+1) \\ = (x+y) + 1 \\ = f(x, y) + 1$$

$$\text{Also } f(x, 0) = x = g(x) = U_1^1(x)$$

$$f(x, y+1) = f(x, y) + 1 \\ = S(f(x, y)) \\ = S(U_3^3(x, y, f(x, y))) \\ = h(x, y, f(x, y))$$

Here $g(x)$ can be defined by $g(x) = x = f(x, 0)$

As $g = U_1^1(x)$, it is an initial function. It is obtained from the initial functions $U_3^3(x, y, z)$ and S by composition, and f by recursion using g and h . Thus f is obtained by applying composition and recursion finite no. of times to initial functions U_1^1, U_3^3 and S . So f is primitive recursive.

Here, $f(2,0) = 2$

(19)

$$\begin{aligned}
 f(2,4) &= S(f(2,3)) \\
 &= S(S(f(2,2))) \\
 &= S(S(S(f(2,1)))) \\
 &= S(S(S(S(f(2,0))))) \\
 &= S(S(S(S(2)))) \\
 &= S(S(S(3))) \\
 &= S(S(4)) \\
 &= S(5) \\
 &= 6
 \end{aligned}$$

Ex: Using recursion, define the multiplication function $*$ given by

$$f(x,y) = x * y.$$

→ As multiplication of two natural numbers is simply repeated addition, f has to be primitive recursive. Here f is a function of two variables. If we want f to be defined by recursion, we need a function g of a single variable and a function h of those variables.

$$f(x,0) = 0$$

$$\begin{aligned}
 \text{and } f(x,y+1) &= x * (y+1) = f(x,y) + x \\
 &= S(f(x,y), x)
 \end{aligned}$$

$$\text{we can write } f(x,0) = 0 = Z(x)$$

$$\text{and } f(x,y+1) = S(U_3^3(x,y,f(x,y)), U_1^3(x,y,f(x,y)))$$

By taking $g = Z$ and h defined by $h(x,y,z) = S(U_3^3(x,y,z), U_1^3(x,y,z)) = f(x,y+1)$, we see that f is defined by recursion.

As g and h are primitive recursive, f is primitive recursive.

(*) Show that $f(x, y) = x - y$ is primitive recursive or proper (20)
 Subtraction is primitive recursive. Hence compute the value of $f(5, 3)$.

→ Proper subtraction is defined as

$$x - y = \begin{cases} x - y & \text{if } x \geq y \\ 0 & \text{if } x < y \end{cases}$$

Hence we use,

$$f(x, y) = x - y$$

$$f(x, 0) = x = g(x) = U_1^1(x)$$

$$\begin{aligned} f(x, y+1) &= x - (y+1) \\ &= x - y - 1 \\ &= f(x, y) - 1 \\ &= p(f(x, y)) \\ &= p(U_3^3(x, y, f(x, y))) \\ &= h(x, y, z) \end{aligned}$$

Hence, $f(x, y)$ is primitive recursive.

$$\begin{aligned} f(5, 3) &= f(5, 2+1) \\ &= f(5, 2) - 1 \\ &= p(f(5, 2)) \\ &= p(p(f(5, 1))) \\ &= p(p(p(f(5, 0)))) \\ &= p(p(p(5))) \\ &= p(p(4)) \\ &= p(3) \\ &= 2 \end{aligned}$$

$\begin{cases} p(x) = x - 1 \\ \text{where } p \rightarrow \text{is predecessor} \\ \text{function} \end{cases}$

(*) Show that $f(x, y) = x^y$ is primitive recursive.

→ $f(x, y) = x^y$

$$f(x, 0) = x^0 = 1 = g(x) = U_1^1(1)$$

$$\begin{aligned} f(x, y+1) &= x^{y+1} = x^y * x \\ &= x * x^y \\ &= x * f(x, y) \\ &= U_3^3(x, y, f(x, y)) \\ &= h(x, y, z) \end{aligned}$$

∴ Hence it is primitive recursive.